Francisco Madaleno

PhD Fellow @ MLSM DTU

Education

Technical University of Denmark (DTU)

PhD Candidate @ **Machine Learning for Smart Mobility (MLSM)**

October 2024 - Present, Copenhagen, Denmark

PhD Topic: Active Learning Causal Abstractions on Simulators for Climate Change Policy Making

Instituto Superior Técnico / MSc Aerospace Engineering

2014 - 2019, Lisbon, Portugal, GPA: 16/20

MSc Thesis: "Flying Tourist Problem: An Integer Linear Programming Approach" as a Research Student at **INESC-ID, Lisbon**. An optimization tool to find the best schedule and route for a multi-city flight trip with real-time data integrated in a web application. Mark: 18/20

Athens Programme 2019: Operations Research in the Airline Industry, **Mines Paristech**
Ten-day intensive course with **AirFrance KLM** Operations Research/Data Science Team.

Summer Course 2018: Experimental Aerodynamics, **University of Southern Denmark**
Optimization project of a wind turbine blade for **LM Wind Power**.

Polytechnic University of Catalonia, UPC - ESEIAAT / Erasmus

2017 - 2018, Barcelona, Spain

Erasmus exchange program: Specialization in Optimization and Operations Research.

Experience

Zalando SE / Applied Scientist

September 2021 - August 2024, Berlin, Germany

Logistics and Algorithms. Developed strategies and algorithms for logistics networks to solve assignment and flow problems. Built and managed micro-service infrastructure to run highly-parallelizable algorithms. Contributed to Zalando's open- and inner-source projects as well as to the company's data science community.

Presented in GOR Workshop 2022: Applied Methods from Mathematical Optimization and Machine Learning for e-commerce. "Mathematical Optimization Meets Machine Learning to Optimize Stock Distribution at Zalando"

Zalando SE / Intern Applied Scientist

April 2021 - September 2021, Berlin, Germany

Pricing and forecasting. Collaborated on a MIP tool that consumes a forecast of over 500k items across 14 markets to recommend discounts over the period of 26 weeks.

Developed a heuristic that defines feasible target ranges as bounds for the MIP tool.

von Karman Institute / INOV Contacto 2020

February 2020 - September 2020, Brussels, Belgium

Participated in the program INOV Contacto 2020 funded by AICEP Portugal, in which I worked for the **von Karman Institute for Fluid Dynamics** as a Research Intern Engineer.

At von Karman, I was responsible for building a computational model and simulation tool to optimize a system of a turbine for the **Safran Group.**

Volunteering

ReDI School of Digital Integration

October 2023 - October 2024, Berlin, Germany

Teacher Assistant in 1. Introduction to Computer Science and 2. Introduction to Machine Learning. Taught Python fundamentals, ML foundational knowledge and assisted students in their first semester final project.

ReDI is a non-profit tech school providing migrants and marginalized locals free and equitable access to digital education.

Publications

Peer-Reviewed:

2026

- Coarsening Causal DAG Models, Oral @ CLeaR 2026,
- Bayesian Hierarchical Invariant Prediction, Poster @ CLeaR 2026,
- Coarsening Linear Non-Gaussian Causal Models with Cycles, Poster @ NeurIPS 2026

Preprints:

2026

- Coarse Causal Graphs for Bayesian Optimization

Oral Presentations

2026

- Contributed Talk at the 5th Conference on Causal Learning and Reasoning (CLeaR) 2026: Coarsening Causal DAG Models

2025

- Contributed Talk at 41st Conference on Uncertainty in Artificial Intelligence Workshop on Causal Abstractions and Representations: Bayesian Hierarchical Invariant Prediction

Conferences Attended

2026

- 5th Conference on Causal Learning and Reasoning (CLeaR) 2026

2025

- 4th CLeaR (Causal Learning and Reasoning) 2025
- 41st Conference on Uncertainty in Artificial Intelligence
- Fast Machine Learning for Science Conference 2025
- EurIPS 2025

Summer Schools

2026

- Oxford Machine Learning Summer School: MLx Representation Learning & Gen AI

2025

- Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2025

Poster

2026

- 5th Conference on Causal Learning and Reasoning (CLeaR) 2026: Coarsening Causal DAG Models
- 5th Conference on Causal Learning and Reasoning (CLeaR) 2026: Bayesian Hierarchical Invariant Prediction

2025

- Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2025: Bayesian Hierarchical Invariant Prediction
- 41st Conference on Uncertainty in Artificial Intelligence @ Workshop on Causal Abstractions and Representations: Bayesian Hierarchical Invariant Prediction
- AI in Science Summit (AIS2025): The APEX project - Artificial Intelligence for Policy Excellence in the Climate Crisis: initial results and outlook
- EurIPS 2025 @ Causality for Impact Workshop: Case Study: Learning Interventionally Consistent Surrogates for Climate Adaptation

2022

- Improving Zalando's Warehouse Efficiency by Relocating Items, International Conference on Operations Research, Karlsruhe, Germany, 2022

Teaching

2026:

- Model-based Machine Learning

Teaching Assistant

2026:

- Model-based Machine Learning
- Advanced Business Analytics

Reviewing

2026:

- The Fortieth Annual Conference on Neural Information Processing Systems NeurIPS 2026),
- Journal of the Royal Statistical Society, Series B,
- npj Urban Sustainability

2025:

- Journal of the Royal Statistical Society, Series B,
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)

Grants / Funding

2026

- Thomas B. Thriges Fond for Research Visit
- Otto Mønsted Fond for Research Visit
- William Demant Fond for Research Visit

Supervision

- Filip Arthur Blaafjell (MSc Thesis) - Causal Structure from Code: An Agentic LLM Pipeline for Causal Structure Extraction from Scientific Simulators
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Community Engagement

2026

- Organized seminar on Spatio-Temporal Machine Learning (Invited Speaker: Gerrit Grossmann (DFKI))
- Presented Coarsening Causal DAG Models at CoCaLa group at the University of Copenhagen
- Participated and organised meetings of the CPHDisco Network (Coordinated by Anne Helby Petersen from the University of Copenhagen)

2025

- Organised seminar on Causal Representation Learning (Invited Speaker: Simon Bing (TU Berlin))
- Participated and organised meetings of the CPHDisco Network (Coordinated by Anne Helby Petersen from the University of Copenhagen)